

data analysis and graphics packages that you will need. You can download additional packages directly from the command prompt in R using the `install.packages` and `library` commands.

```
> install.packages("package name")
> library(package name)
```

Learning resources

A huge benefit to using R is the built-in help function. For example, to find out more about boxplots, open R and type “`?boxplot`” into the command console. You will get details about boxplot usage and arguments to pass along with examples and references. A great way to learn R is through a short interactive learning resource called Swirl (<http://swirlstats.com>) that allows you to learn about basic R operations through the R console.

To install Swirl, open R > type `install.packages("swirl")` > `library("swirl")` > `swirl()`

A longer way to learn R applied to data Science is to work through the nine courses in the Data Science specialisation offered by John Hopkins University on Coursera.

Go to <https://www.coursera.org> > Click Specialisations > click Data Science

If you are looking for a more basic course, try the Data Analysis and statistical inference course by Duke University on Coursera, or the Foundations of Data Analysis course by the University of Texas on EdX.

Some of the best books to use to learn R are:

- ◆ Victor A. Bloomfield. Using R for Numerical Analysis in Science and Engineering. Chapman & Hall/CRC, 2014. ISBN 978-1439884485
- ◆ Sarah Stowell. Using R for Statistics. Apress, 2014. ISBN 978-1484201404.
- ◆ Steven Murray. Learn R in a Day. SJ Murray, 2013.
- ◆ Robert J Knell. Introductory R: A Beginner's Guide to Data Visualisation and Analysis using R. March 2013. ISBN 978-0-9575971-0-5.
- ◆ J.C. Nash. Nonlinear Parameter Optimization Using R Tools. Wiley, 2014. ISBN 9781118883969. 